

Statistical Quality Control

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SQC → SQC is the term used to describe the set of statistical tools used by quality professionals.

↳ it was pioneered by Walter A. Shewhart (1920)s

Descriptive Statistics

Mean → measure of central tendency

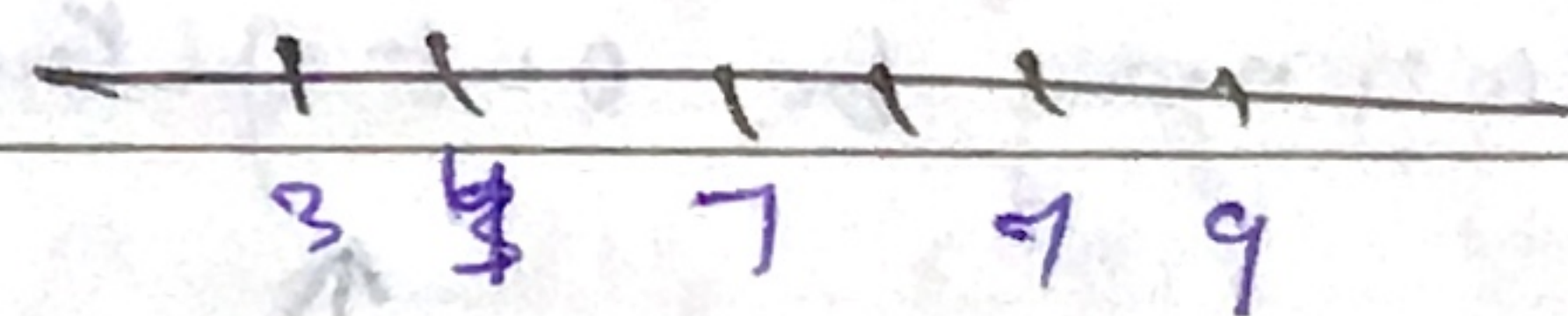
Range → diff. b/w largest and smallest observation in a set of data.

Standard deviation → measure of the amount of data dispersion around mean.

Mean

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Range



Range
 $9 - 3 = 6$

Std. deviation

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{(n-1)}}$$

It measures the extent to which these values are scattered around the central mean.

Good Write

process variation

The natural or inherent variability that occurs in any product manufacturing process

It represents the difference in output of a process due to various factors.

① Common Cause Variation (Natural variation)

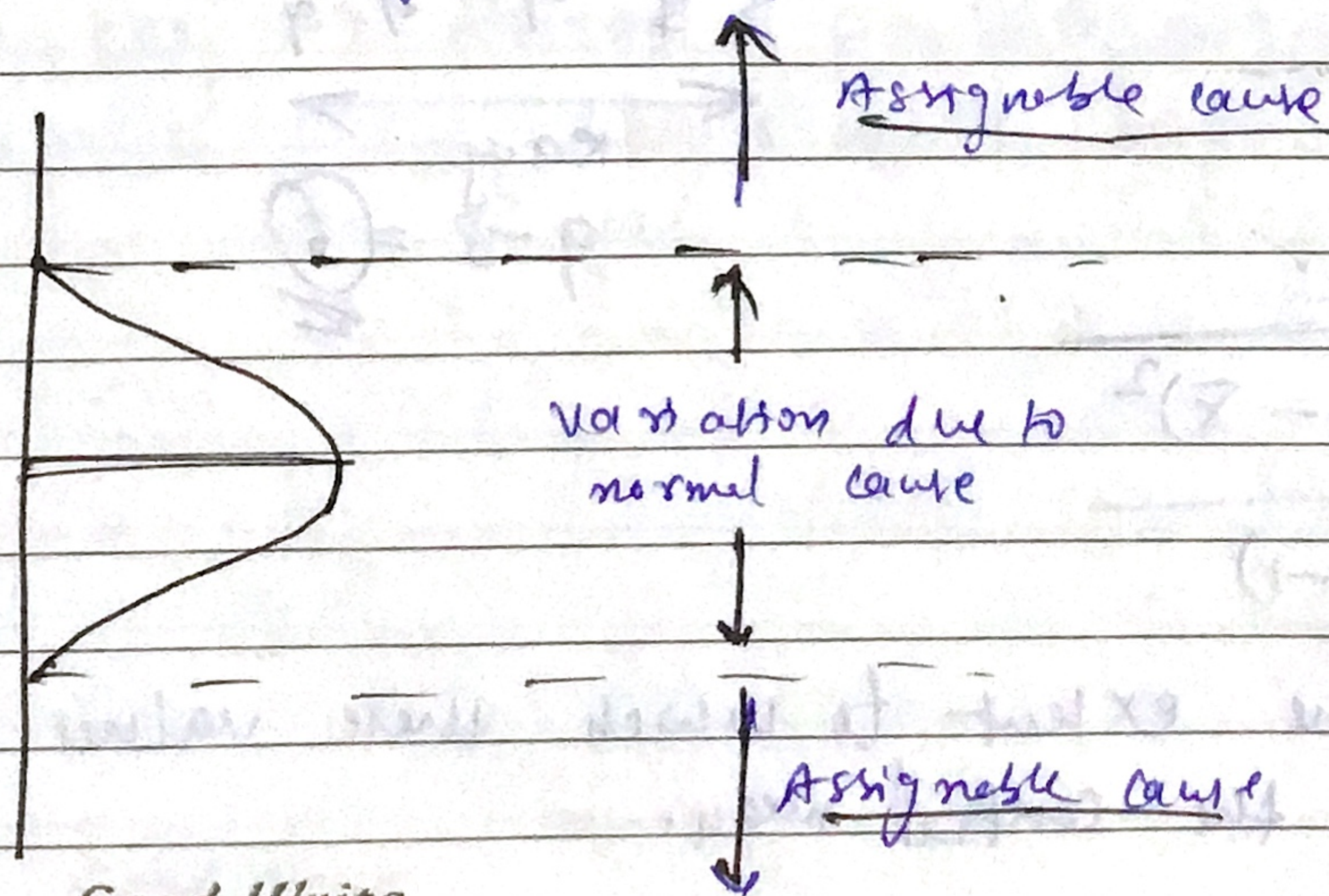
occur due to environment error
functional but inevitable

Not caused by machine error
these are expected error

② Special Cause Variation (Assignable variation)

caused by special, identifiable factors
machine malfunction
Human Error

Use of wrong material
Process may be out of control

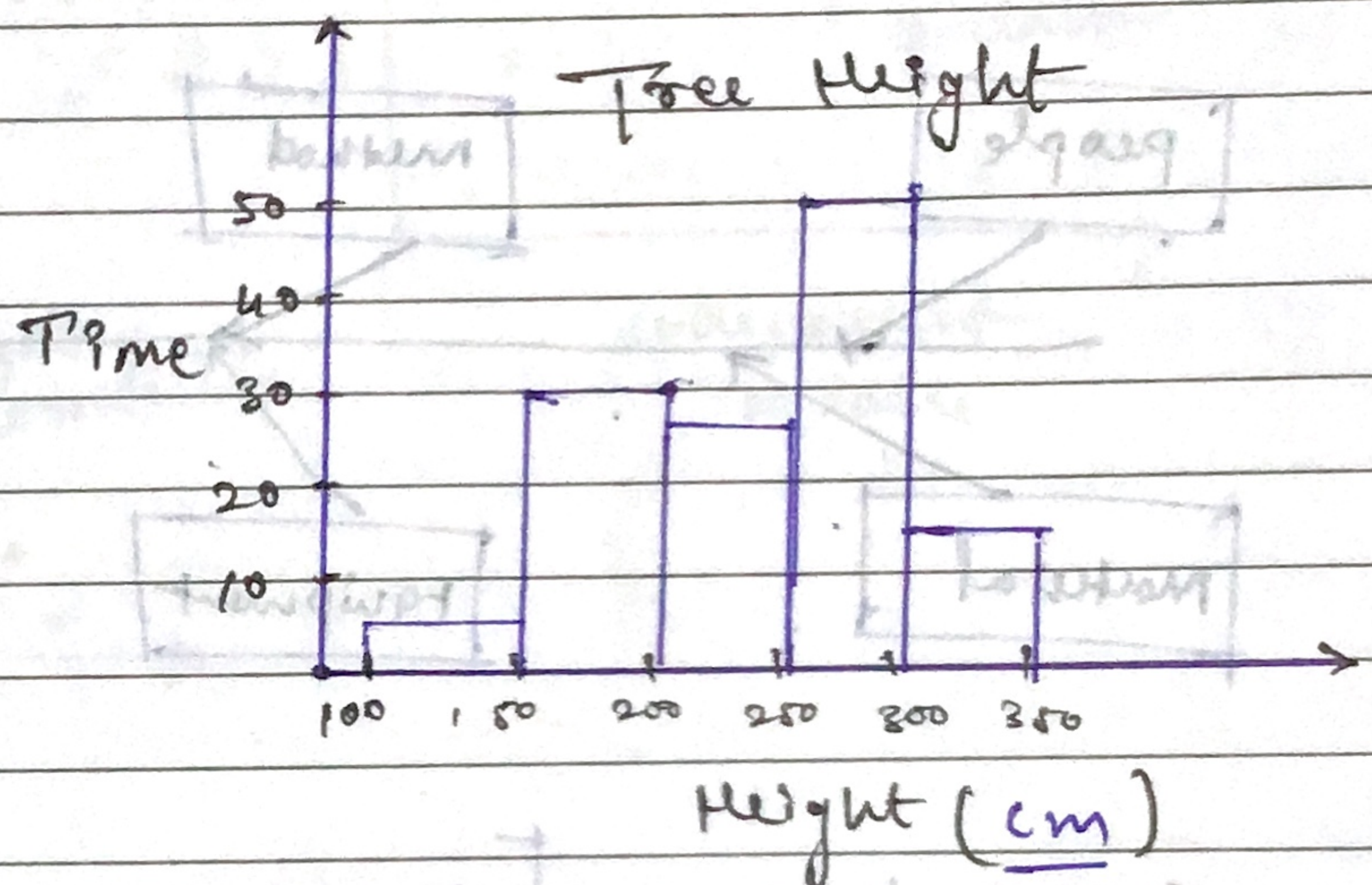


Histogram

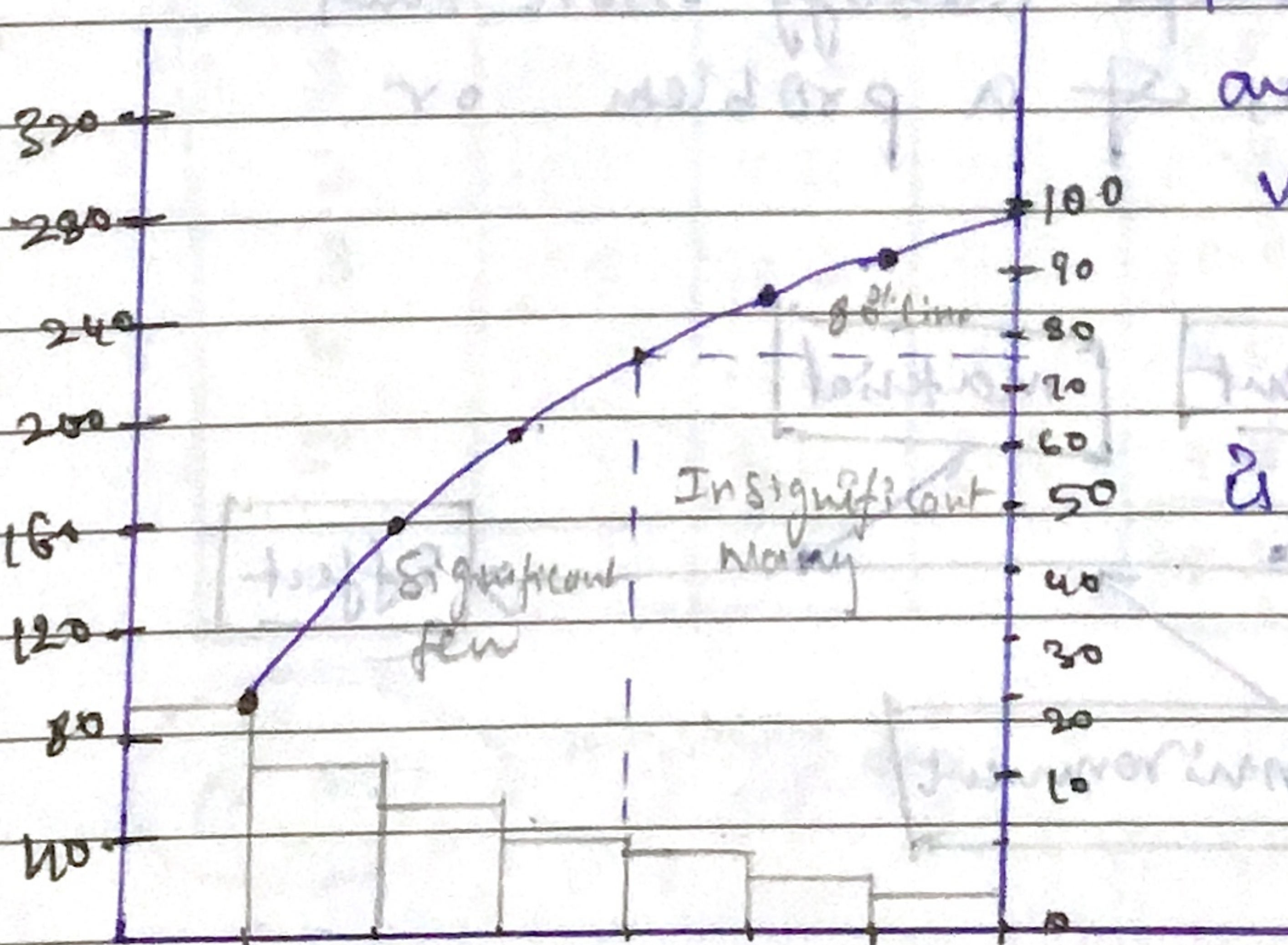
It is a graphical representation of the distribution of numerical data.

It is an estimate of the probability distribution of a continuous variable (quantitative variable).

Introduced by Karl Pearson



Pareto chart



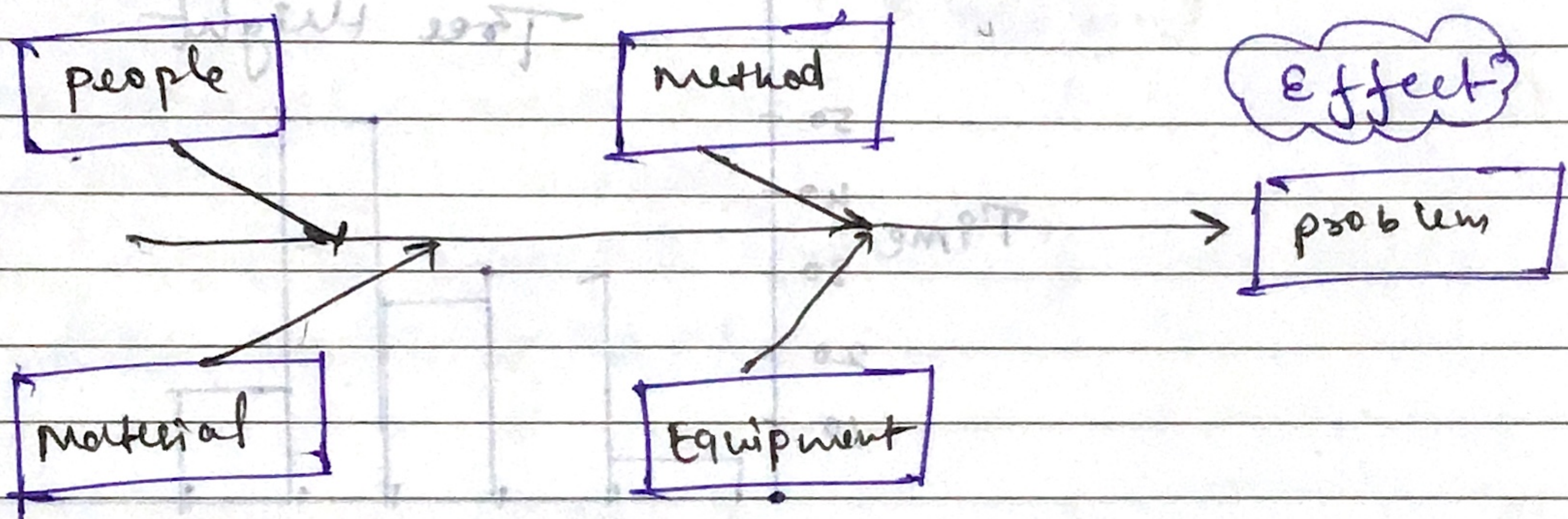
These charts contain both bars and lines graph, where individual values are represented in descending order by bars and the cumulative total (%) is shown by line.

80% of consequences come from 20% of the causes.

Fish bone

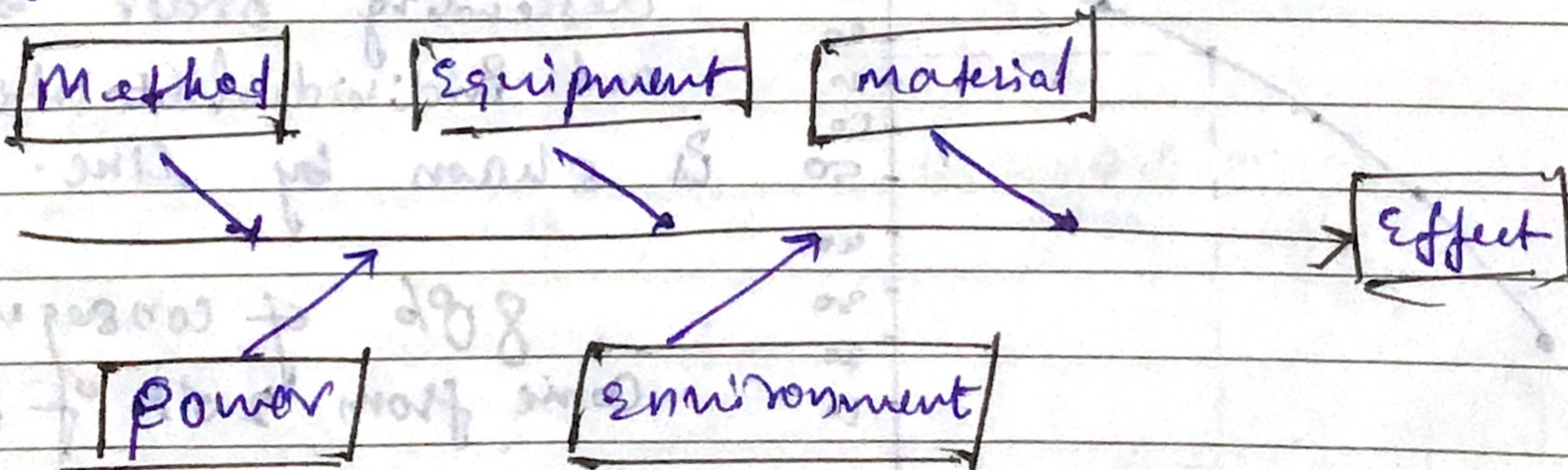
This diagram is useful to organize ideas and to identify relationships.

The usual approach is to consider four problem areas → method, material, Equipment and person/manpower.



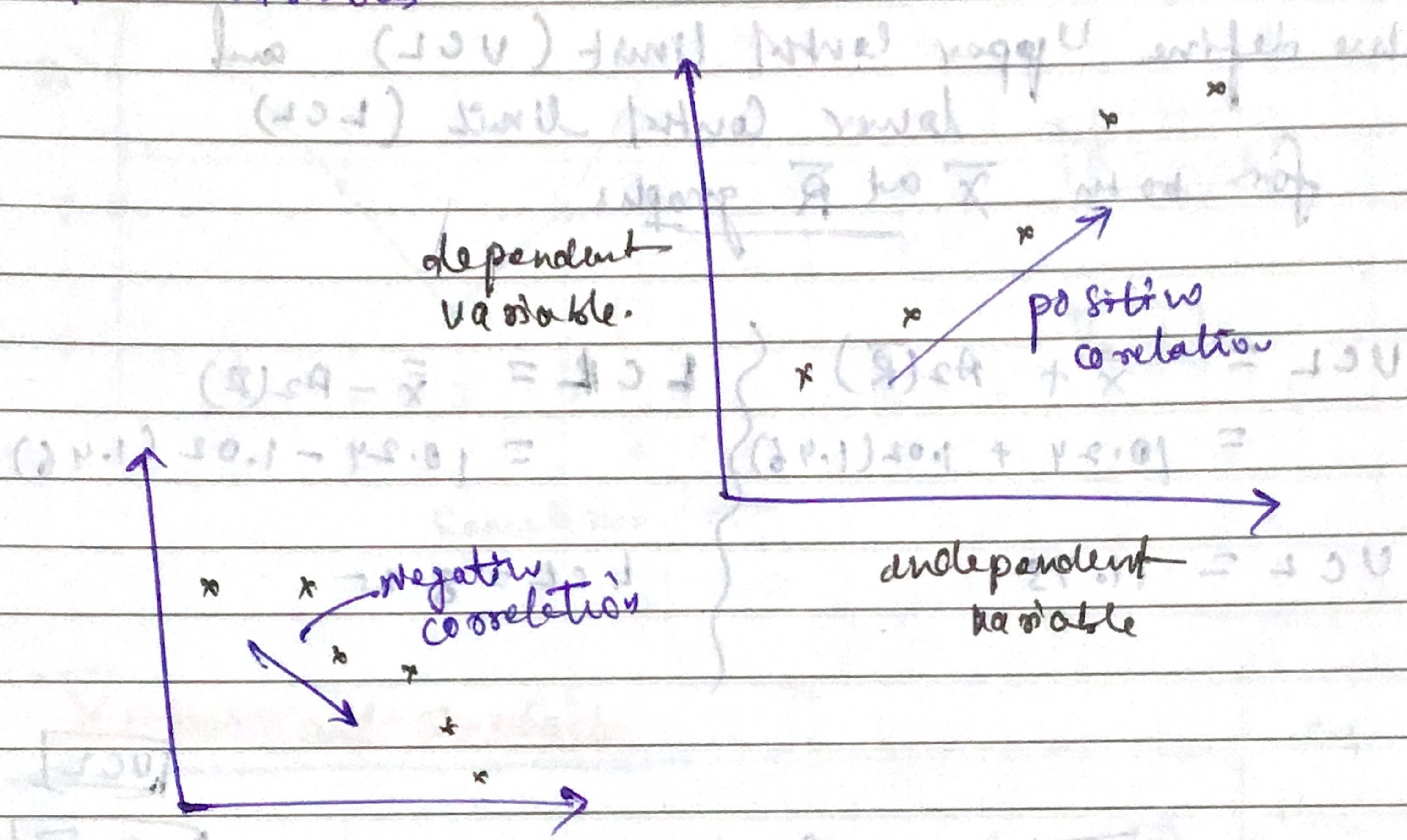
Cause & effect [Also known as]

Graphical tool that helps identify, sort and display possible causes of a problem or quality characteristics.



Scatter diagram

If the graph or plot that shows relation b/w two variables.



X-bar & R-bar charts

Sample	Day 1	Day 2	Day 3	$\bar{X}$	R
1	11.1	9.2	11.3	10.5	2.1
2	10.1	11.2	9.9	10.4	1.3
3	9.8	10.2	9.9	10.0	0.4
4	11.3	10.1	10.1	10.5	1.2
5	11.2	9.4	8.9	9.8	2.3
				10.24	

$$\bar{R} = \frac{2.1 + 1.3 + 0.4 + 1.2 + 2.3}{5}$$

$$\bar{R} = 1.46$$

$$\bar{\bar{X}} = \text{mean of all mean}$$

$$= \frac{10.5 + 10.4 + 10.0 + 10.5 + 9.8}{5}$$

$$\bar{\bar{X}} = 10.24$$

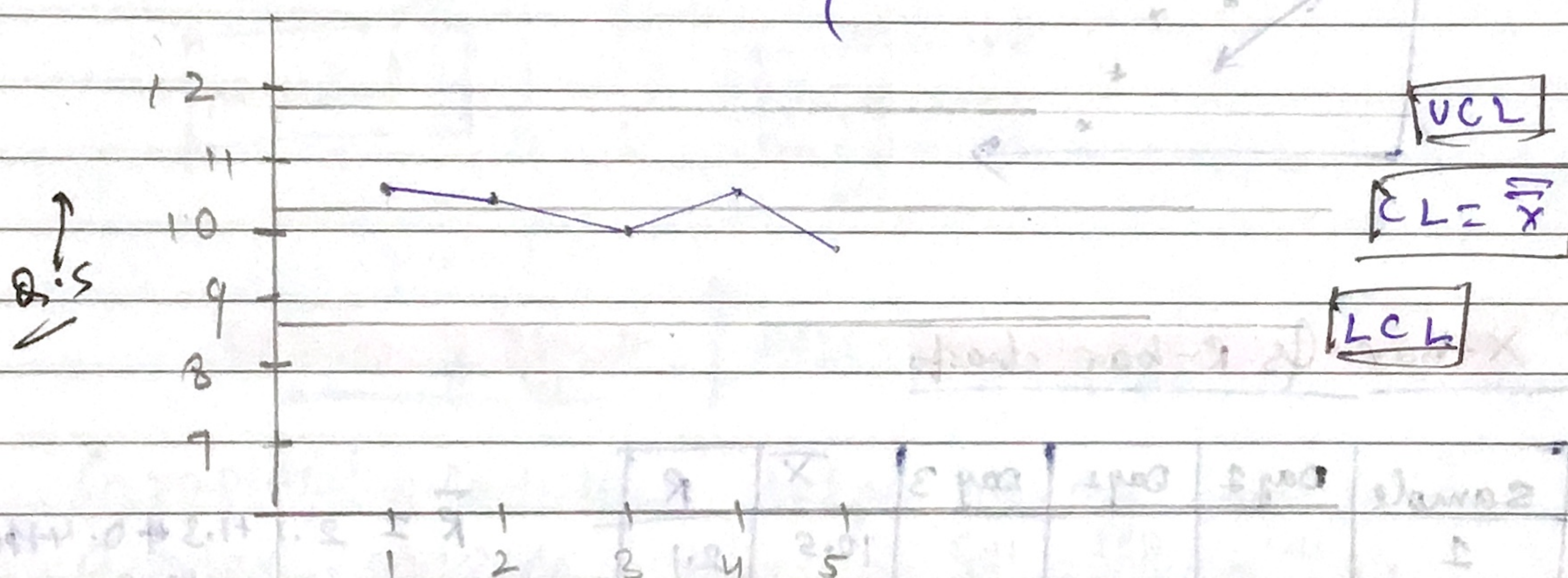
$N = 3$ ← no. of days

for X-bam

$$A_2 = 1.02$$

We define Upper Control Limit (UCL) and lower Control limit (LCL) for both $\bar{X}$ and $\bar{R}$ graphs.

$$\begin{aligned} UCL &= \bar{\bar{x}} + A_2(\bar{R}) \\ &= 10.24 + 1.02(1.46) \\ UCL &= 11.7392 \end{aligned} \quad \left\{ \begin{aligned} LCL &= \bar{\bar{x}} - A_2(\bar{R}) \\ &= 10.24 - 1.02(1.46) \\ LCL &= 8.7508 \end{aligned} \right.$$



for R-bar

$$D_3 = 0$$

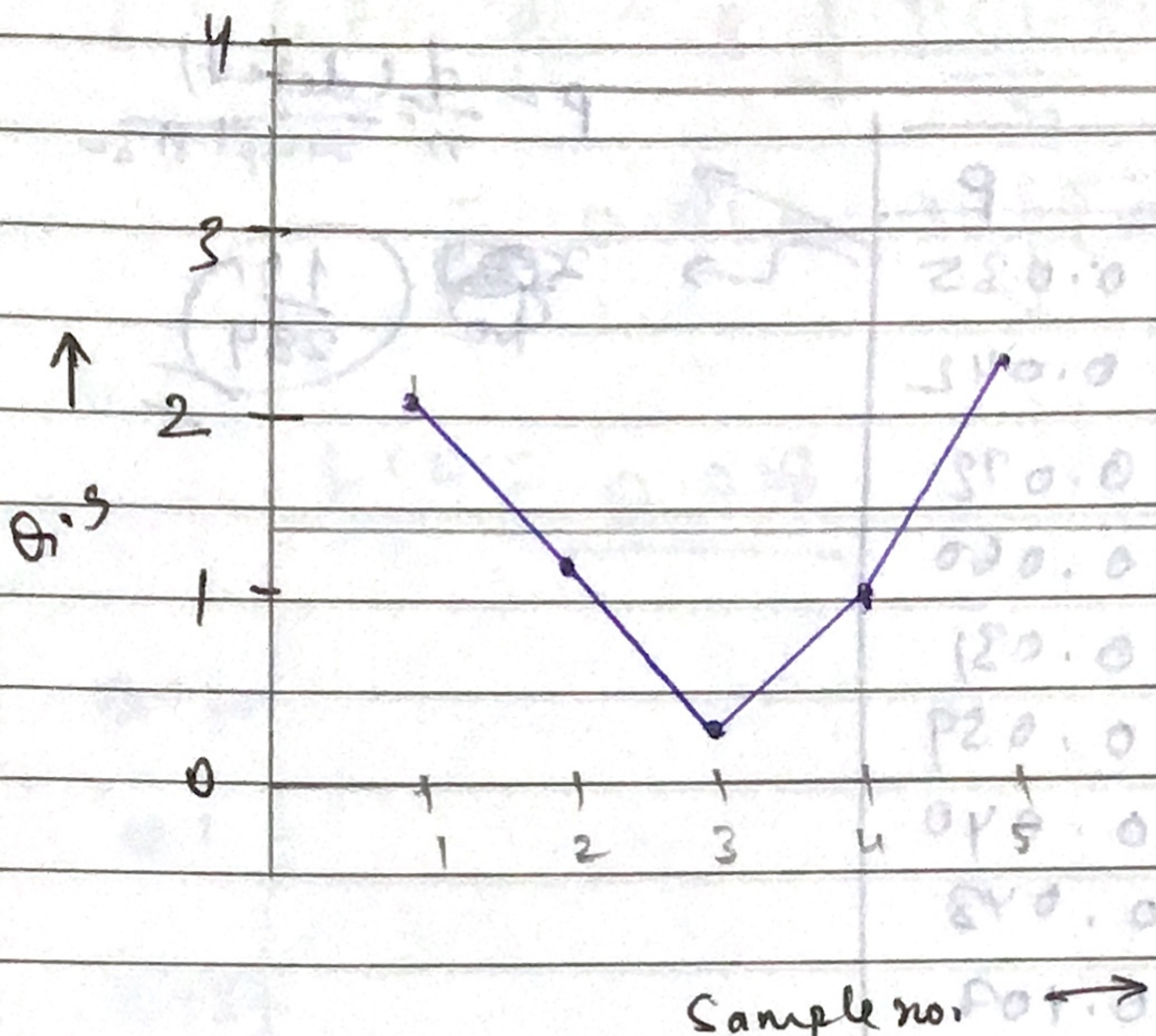
$$Du = 2.57$$

$$\overleftarrow{R} = 1046.$$

$$UCL = D_4(\bar{R}) = 2.57(1.46)$$

$$LCL = D_3(\bar{R}) = 0 \quad (1.46)$$

$$LCU = 0$$



X-bar and S-chart

Everything is same as of $(\bar{X} - R)$ chart

just
$$S = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n-1}}$$

for $\bar{X}$ - chart

$$UCL = \bar{X} + A_2(\bar{R})$$

~~LCL~~

$$LCL = \bar{X} - A_2(\bar{R})$$

for S-chart

$$UCL = B_4 \bar{S}$$

$$LCL = B_3 \bar{S}$$

P-chart

Defect

$$p = \frac{d}{n} \text{ (defect)} \quad \text{Sample size}$$

Day	Non conformers	Sample size	P
1	10	284	0.035
2	12	287	0.042
3	13	180	0.072
4	15	250	0.060
5	9	294	0.031
6	12	203	0.059
7	11	277	0.040
8	12	249	0.048
9	13	120	0.108
10	12	165	0.073
	119	2249	

$$\bar{p} = \frac{119}{2249}$$

$$\bar{p} = 0.053$$

$$\bar{n} = \frac{2249}{10}$$

$$\bar{n} = 224.9$$

$$UCL = \bar{p} + 3 \sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}}$$

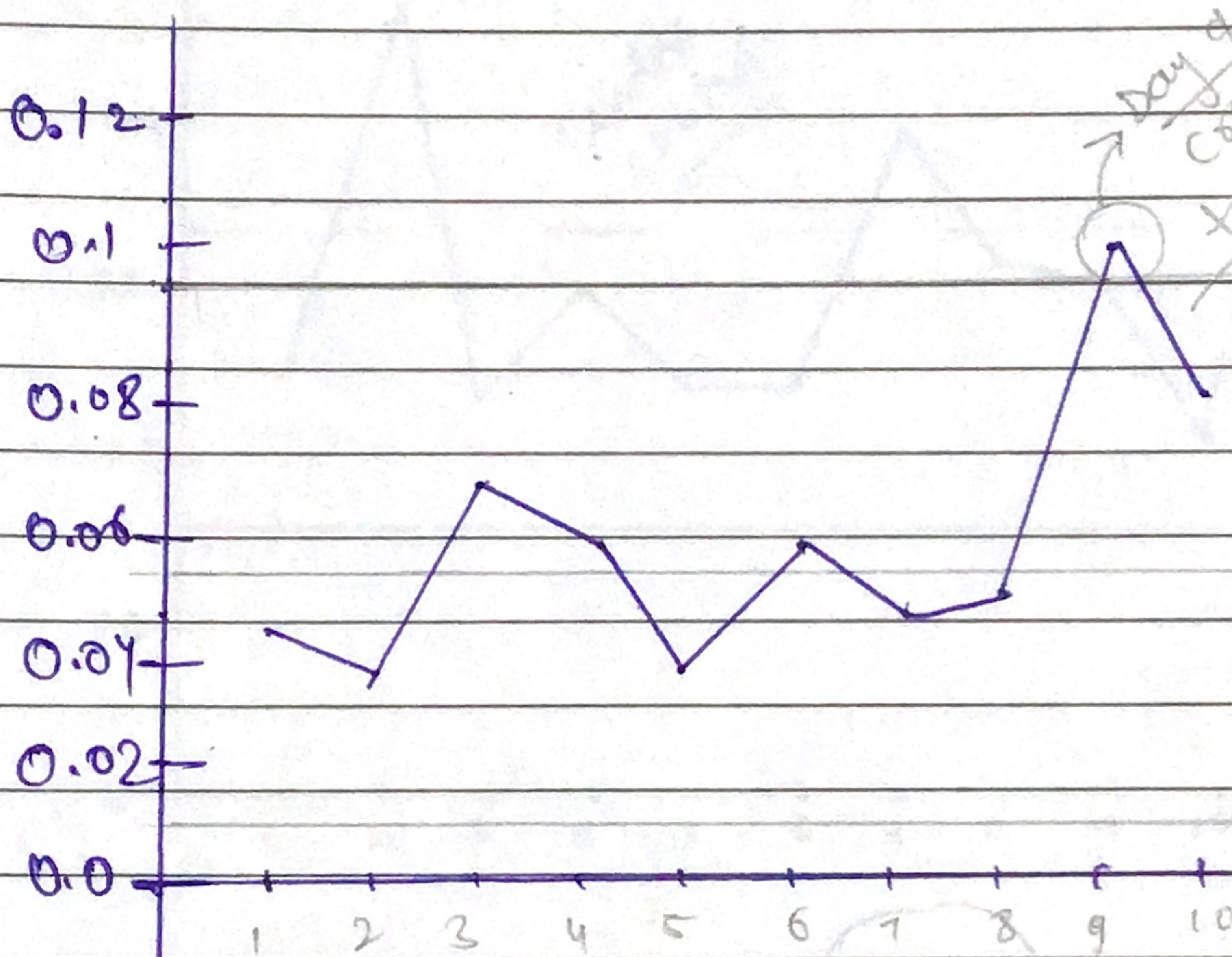
$$= 0.053 + 3 \sqrt{\frac{(0.053)(0.947)}{224.9}}$$

$$UCL = 0.098$$

$$LCL = \bar{p} - 3 \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

$$= 0.053 - 3 \sqrt{\frac{(0.053)(0.947)}{224.9}}$$

$$LCL = \underline{0.028}$$



C-chart

Days	Defects
1	3
2	5
3	1
4	1
5	15
6	15
10	15
Total →	86

$$\bar{c} = \frac{86}{10}$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}}$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}}$$

If $LCL < 0$ then put $LCL = 0$

Np-chart

Sample	No. of Defects
1	25
2	47
3	23
4	30
5	24
6	24
7	39
8	32
9	35
10	22
11	45
12	40
<u>Total</u>	<u>396</u>

$$n\bar{p} = \frac{\sum np}{n} = \frac{396}{12} = 33$$

$$p = \frac{n\bar{p}}{n} = \frac{33}{250} \leftarrow \text{inspected items}$$

$$CL = n\bar{p}$$

$$UCL = n\bar{p} + 3\sqrt{n\bar{p}(1-p)}$$

$$UCL = 33 + 3\sqrt{33\left(1 - \frac{33}{250}\right)}$$

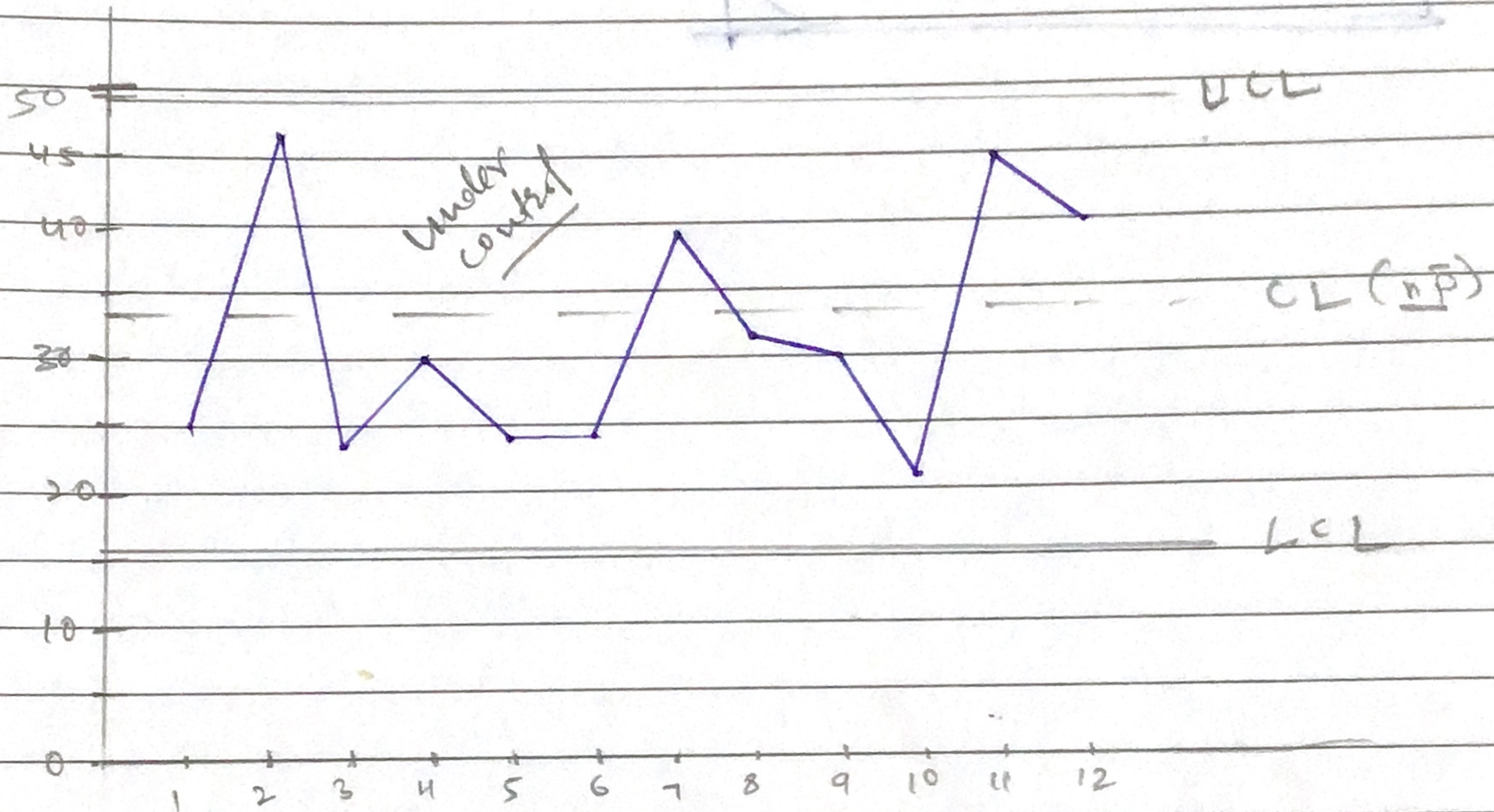
$$UCL = 49.05$$

$$LCL = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$$

$$= 33 - 16.05$$

$$LCL = 16.95$$

$$(n\bar{p} - 4.34) = 29$$



process capability

Compares actual measurement data from quality control processes to the design specification limits by using capability indices like C_p & C_{pk}

$$USL = 14$$

$$LSL = 10$$

$$\text{Mean } (\bar{x}) = 12$$

$$\bar{x} - \bar{bar} = 11.6$$

$$R - \bar{bar} = 3$$

Good Write

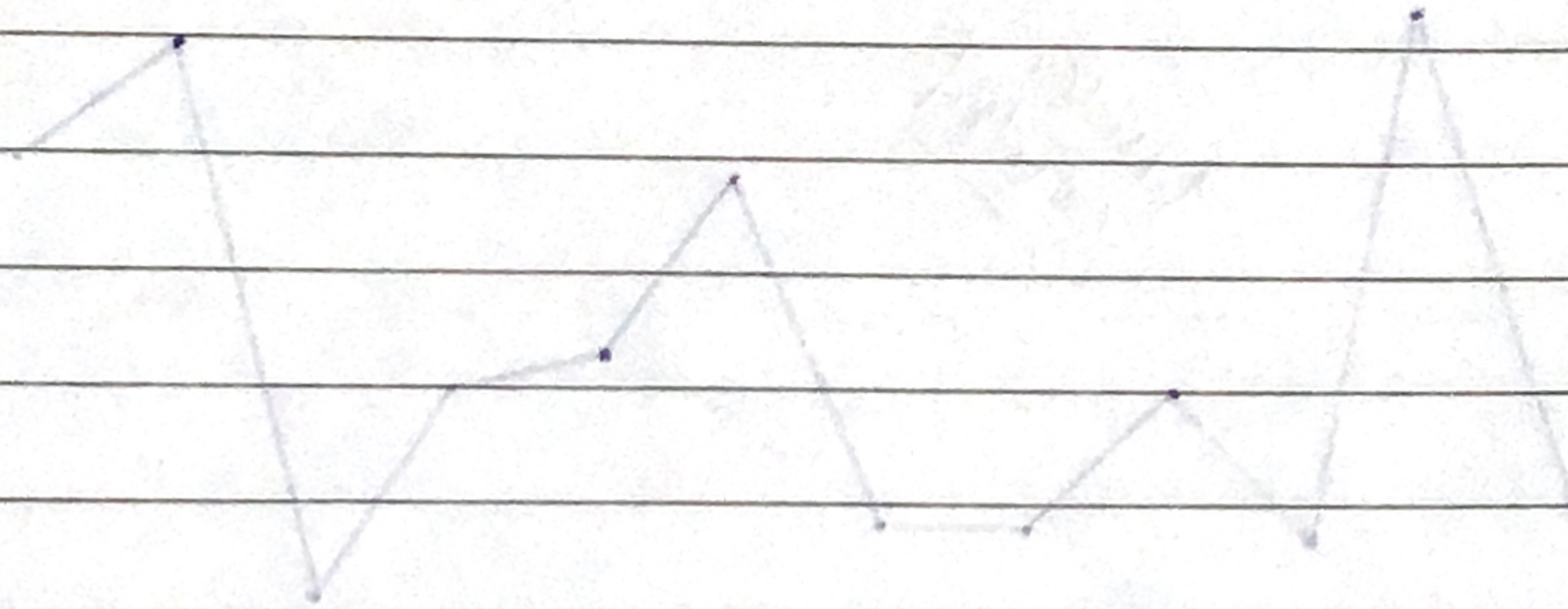
$$D_2 = 2.326$$

$$\sigma = \frac{R - \bar{bar}}{D_2} = \frac{3}{2.326}$$

$$C_p = \frac{USL - LSL}{6\sigma}$$

↑ C_p means, less items will be rejected

$$C_{pk} = \frac{UCL - \bar{x} - \text{bar}}{3\sigma}$$



$$C_{pk} = \frac{UCL - \bar{x} - \text{bar}}{3\sigma}$$

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